5. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S0 \mid A \; A \rightarrow 1A \mid \varepsilon$

Code:

```
#include <stdio.h>
#include <string.h>
int main() {
    char s[100];
    int i = 0, j, c = 0;
    printf("Enter string: ");
    scanf("%s", s);
    j = strlen(s) - 1;
    while (i < j && s[i] == '0' && s[j] == '0') {
        c++;
        i++;
        j--;
    }
    while (i <= j) {
        if (s[i] != '1') {
            printf("String does not belong to CFG");
            return 0;
        }
        i++;
    }
   printf("String belongs to CFG");
    return 0;
}
```

Output: